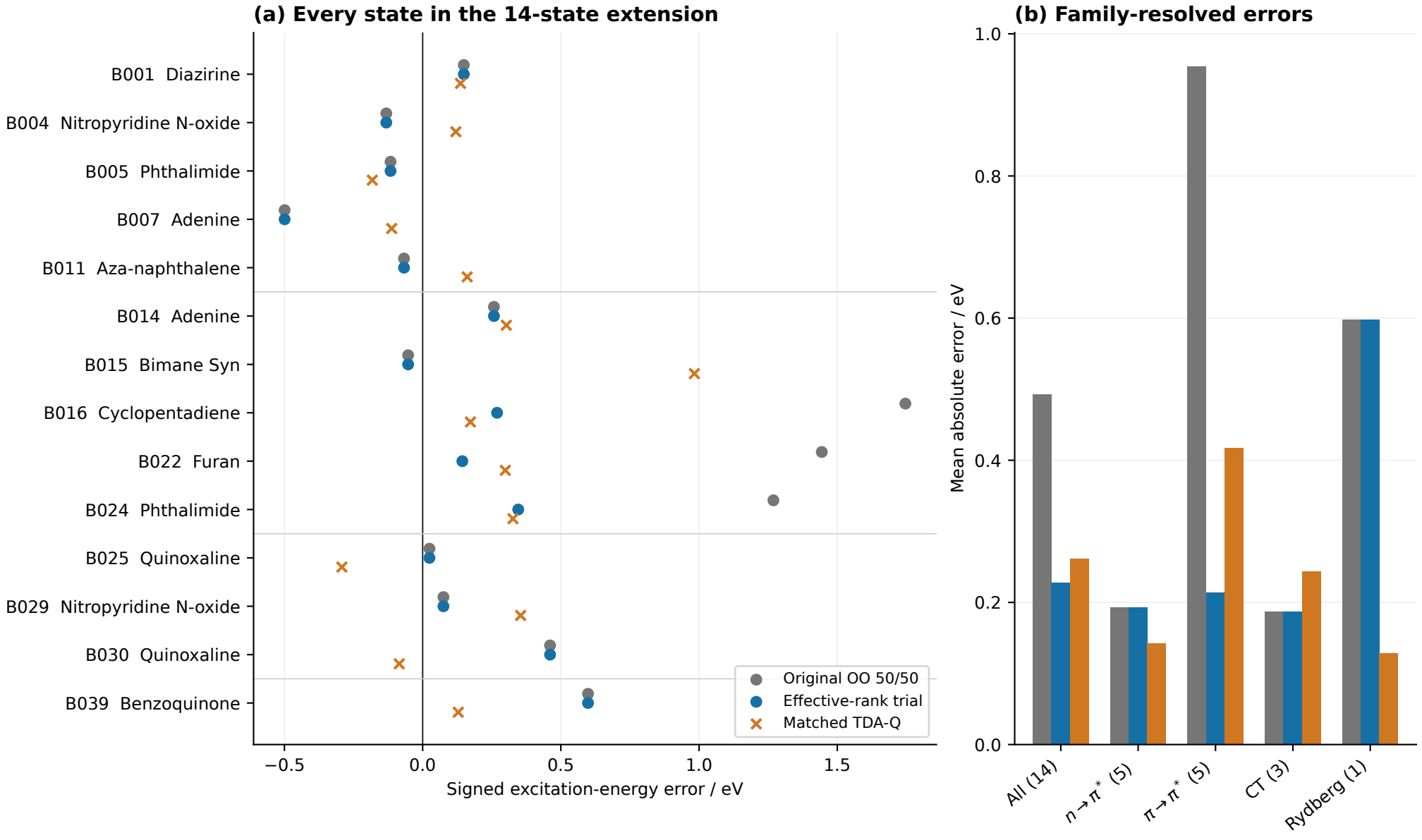
(a) Every state in the 14-state extension
B001 Diazirine
B004 Nitropyridine N-oxide
B005 Phthalimide
B007 Adenine
B011 Aza-naphthalene
B014 Adenine
B015 Bimane Syn
B016 Cyclopentadiene
B022 Furan
B024 Phthalimide
B025 Quinoxaline
B029 Nitropyridine N-oxide
B030 Quinoxaline
B039 Benzoquinone
−0.5
0.0
0.5
1.0
1.5
Signed excitation-energy error / eV
Original OO 50/50
Effective-rank trial
Matched TDA-Q
(b) Family-resolved errors
Mean absolute error / eV
0.0
0.2
0.4
0.6
0.8
1.0
All (14)
n→π* (5)
π→π* (5)
CT (3)
Rydberg (1)